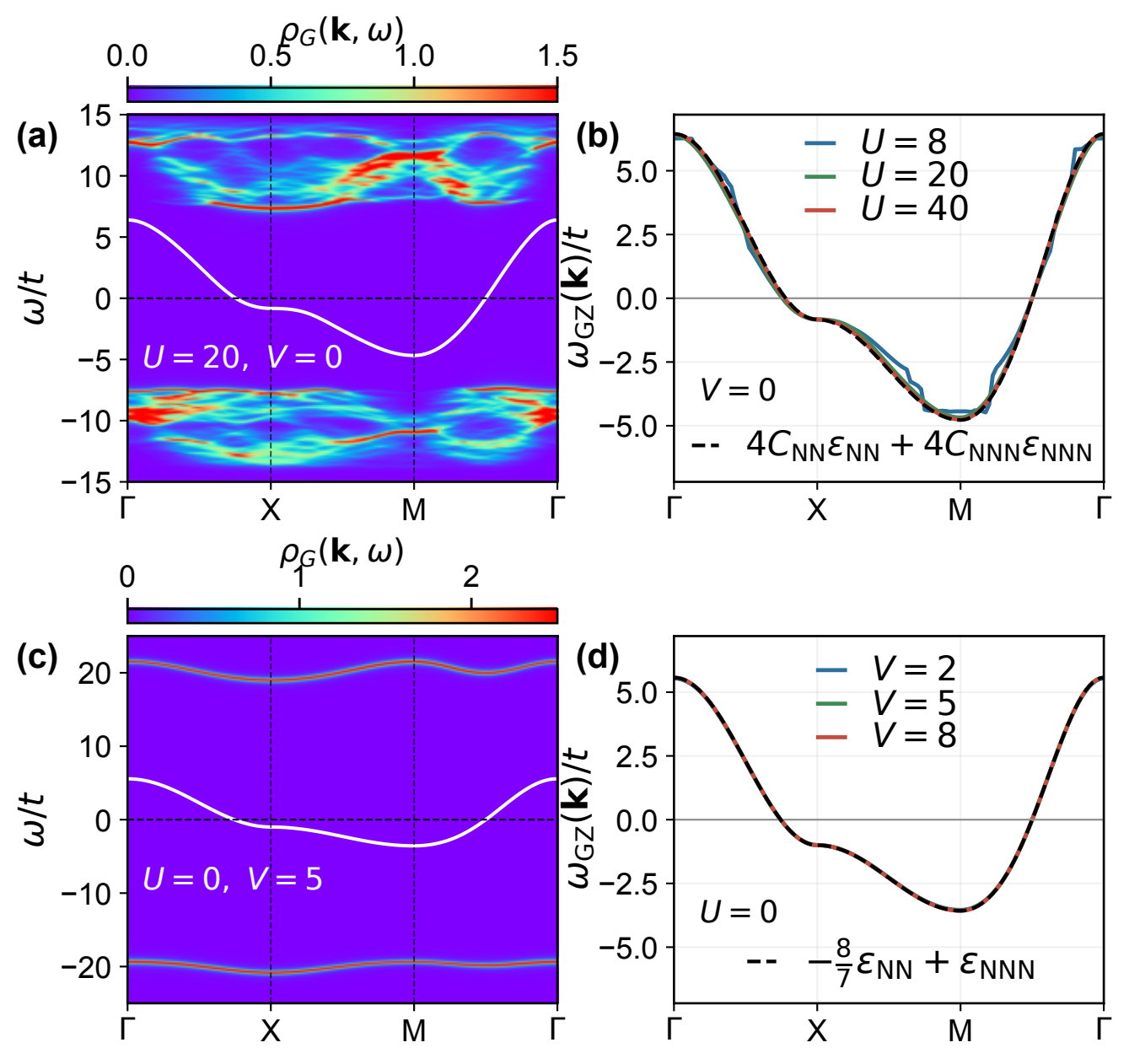

(a)
$\rho_G(\mathbf{k},\omega)$
0.0
0.5
1.0
1.5
15
10
5
0
−5
−10
−15
ω/t
$U = 20,\ V = 0$
Γ
X
M
Γ
(b)
$U = 8$
$U = 20$
$U = 40$
5.0
2.5
0.0
−2.5
−5.0
$\omega_{GZ}(\mathbf{k})/t$
$V = 0$
-- $4C_{NN}\varepsilon_{NN} + 4C_{NNN}\varepsilon_{NNN}$
Γ
X
M
Γ
(c)
$\rho_G(\mathbf{k},\omega)$
0
1
2
20
10
0
−10
−20
ω/t
$U = 0,\ V = 5$
Γ
X
M
Γ
(d)
$V = 2$
$V = 5$
$V = 8$
5.0
2.5
0.0
−2.5
−5.0
$\omega_{GZ}(\mathbf{k})/t$
$U = 0$
-- $-\frac{8}{7}\varepsilon_{NN} + \varepsilon_{NNN}$
Γ
X
M
Γ